INSTRUCTOR PACKET

EBIKE101 Day 3

Motors, Controllers, Throttle/PAS, and Signal Flow

Evidence first. Motion last.

Morning-of launch

- Print one student packet per student.
- Display the matching 20-slide deck.
- Stage one visible e-bike or conversion bike with battery off or disconnected.
- Put SIGNAL PATH and POWER PATH on the board in two colors.
- Stage loose components, removable labels, basic tools, and a multimeter for instructor use.

Learning targets

- Separate signal and power paths.
- Describe motor and controller jobs.
- Explain throttle, PAS, display, and brake-cutoff inputs.
- Trace battery to controller to motor.
- Build an ordered no-assist diagnostic plan.

Core rule

Power off first. Trace before testing. Control motion before adding power.

Five-hour run of show

Time	Block	Instructor move
10:00-10:10	Landing	Attendance and Day 1 safety review
10:10-10:30	Quiz	Check safe thinking; reteach before hands-on work
10:30-10:55	Two paths	Separate commands from working energy
10:55-11:20	Motor	Torque, hub-drive, and mid-drive
11:20-11:45	Controller	Inputs, decision, and controlled output
11:45-12:00	Safety inputs	Throttle/PAS requests; brake cutoff overrides
12:00-12:30	Lunch	Bike stable and power off
12:30-1:05	Lab A	Identify and tag components
1:05-1:45	Lab B	Draw and compare both paths
1:45-2:00	Break	Reset labels and tools
2:00-2:35	Lab C	Trace one no-assist scenario
2:35-2:55	Debrief	Defend first safe test with evidence
2:55-3:00	Close	Preview battery and charging safety

Teaching sequence

1. Power path

Battery -> protection -> controller -> motor -> mechanical rotation

2. Signal path

Throttle, PAS, brakes, and display -> controller decision

3. Controller

The controller is where the paths meet. It reads requests and safety conditions, then controls energy sent to the motor.

4. Diagnostic order

- Make safe
- Confirm the complaint
- Trace the path
- Test one checkpoint at a time
- Record the result

Stop-work triggers

- Unexpected motor or wheel motion
- Unstable bike or driven wheel
- Battery heat, swelling, odor, damage, wetness, or unknown state
- Forced, damaged, or mismatched connector
- Exposed conductor or uncertain polarity
- Wiring near moving parts, sharp edges, heat, or pinch points
- Attempt to bypass a cutoff or test by random connection

Lab setup and fault cards

Use an actual known-safe condition or a paper fault card. Do not create an energized fault. Good scenarios:

- Battery switch is off or system wake-up step is missing.
- Display mode or assist level prevents assist.
- Brake cutoff is active in the scenario.
- A low-voltage signal connector is visually identified as unseated while power remains off.
- PAS alignment or sensor gap needs inspection.
- Motor cable routing or connector condition requires instructor review.

Backup

Use labeled component cards if no complete bike is available. Teams build both paths, receive one fault card, and defend their checkpoint sequence without applying power.

Morning Quiz — Answer Key

Use with the student quiz. Correct answers and teaching rationales are listed below.

#	Correct answer	Why
1	B. Make the system safe and power it off	Make the system safe before tracing it.
2	C. Read inputs and manage power sent to the motor	The controller reads rider and safety inputs, then controls motor power.
3	A. Converts electrical energy into mechanical rotation	The motor turns controlled electrical energy into mechanical rotation.
4	A. A direct request for motor assist	A throttle sends a direct assist request to the controller.
5	C. Pedaling input	PAS detects pedaling activity or cadence input.
6	B. Tell the controller to stop motor drive	A safety input should override a go input.
7	A. Battery -> protection -> controller -> motor	The high-energy path runs from battery through protection and controller to motor.
8	C. Throttle, PAS, brakes, and display	These devices provide rider, mode, or safety signals.
9	B. Signal carries a command; power carries energy	A command and the energy that performs it are different paths.
10	C. Make safe and confirm the complaint	Confirm the condition safely before changing the system.
11	B. Identify and verify it before connection	Unknown connectors must be identified, not forced or guessed.
12	A. Clear of moving parts, sharp edges, heat, and pinch points	Safe routing prevents abrasion, snagging, heat damage, and shorts.
13	C. Stable with the driven wheel and area controlled	Stability and area control prevent unexpected motion from becoming injury.
14	B. At the wheel hub	A hub motor is built into or drives the wheel hub.
15	A. Through the bicycle drivetrain	A mid-drive adds torque through the chain and bicycle drivetrain.

#	Correct answer	Why
16	C. A recorded checkpoint and result	Documentation creates repeatable evidence.
17	B. Stop-work and make the system safe	Unexpected motion is a stop-work trigger.
18	A. Shape, keying, pin count, labels, and routing	Multiple identifiers are stronger evidence than color alone.
19	C. Instructor approval, stable bike, clear area, and controlled motion	A powered test requires control, approval, and a clear area.
20	B. Trace the system instead of guessing at parts	Ordered tracing isolates the problem without random part swapping.

Lab scoring

Category	3 - Job-ready	2 - Acceptable	1 - Needs rework	0 - Unsafe
Safety	Holds power-off boundary	Minor reminder	Repeated reminders	Unsafe action
Identification	All core parts correct	One minor miss	Several errors	Cannot identify path
Diagram	Both paths clear	Mostly correct	Paths mixed	No usable diagram
Diagnosis	Ordered evidence plan	Usable basic plan	Guessing remains	Random connection or bypass
Documentation	Clear checkpoints	Basic notes	Incomplete notes	No record

Successful completion

Students can name the path, mark the checkpoints, explain the first safe test, and stop when evidence does not support the next step.

Finish early

Give teams a second paper fault card and require them to identify which checkpoint changes and which checkpoints stay the same.